ABUBAKAR SADDIQUE

AI Engineer | Generative AI, RAG & Agentic AI Systems
Lahore, Pakistan | +92-308-4555062 | abubakarsaddique3434@gmail.com
github.com/abubakarsaddique22 | linkedin.com/in/abubakrsaddique | Portfolio

PROFESSIONAL SUMMARY

AI Engineer with 6+ months of professional experience in data science and Generative AI, plus freelance client work. Built and deployed a production-style RAG system on AWS with hybrid search, cross-encoder reranking, role-based access control, guardrails, LLM evaluation and CI/CD with automatic rollback. Currently building a WhatsApp AI customer-support bot for a live Shopify store. Seeking a Junior / Associate AI Engineer role.

TECHNICAL SKILLS

Generative AI & LLMs: LLMs, RAG, Hybrid Search (BM25 + Dense, RRF), Cross-Encoder Reranking, Query Rewriting, AI Agents, Multi-Agent Systems, Prompt Engineering, LoRA, QLoRA
Frameworks & Vector DBs: LangChain, LangGraph, CrewAI, Agno, Hugging Face, Sentence Transformers, Groq, Ollama, Qdrant, Pinecone
LLM Evaluation & Safety: DeepEval, RAGAS, LLM-as-a-Judge, LangSmith Tracing, Guardrails, PII Masking (Presidio), Red-Team Testing
Backend & Databases: Python, SQL, FastAPI, REST APIs, PostgreSQL, Redis, SQLAlchemy, JWT Auth, Pydantic, Pytest
MLOps & Cloud: Docker, GitHub Actions (CI/CD), AWS (EC2, ECR, S3, IAM, SSM), Prometheus, Grafana, MLflow, DVC, Git
ML & Data: Scikit-learn, XGBoost, PyTorch, TensorFlow, NLP, Pandas, NumPy, Streamlit, Gradio, Power BI, Plotly

PROFESSIONAL EXPERIENCE

AI Engineer (Freelance) | Tussles Store Oct 2026 – Present | Lahore
WhatsApp AI Customer Support Bot for a Shopify store: Python, FastAPI, Meta WhatsApp Cloud API, Shopify API, Groq LLM + Whisper, RAG, PostgreSQL, Docker, LangSmith, GitHub Actions

- Building a WhatsApp AI support bot that answers customer product, price and store-policy questions in Roman Urdu and English, using live Shopify catalog data (synced via webhooks) and a RAG FAQ knowledge base.
- Added voice-note understanding (Groq Whisper large-v3), typo-tolerant product search, price filters, human handoff alerts, and prompt-injection detection with a cool-down for repeated attacks.
- Backed by 2,000+ automated tests, LangSmith evaluation and GitHub Actions CI; built onboarding for the store's existing WhatsApp Business number via Meta Embedded Signup (Coexistence); AWS EC2 deployment in progress.

Data Science & GenAI Intern | Salaar Technologies Pvt. Ltd. Oct 2025 – Apr 2026 | Lahore

- Developed Python data pipelines for web scraping, data cleaning, feature engineering and structured data preparation for downstream machine learning workflows.
- Built and evaluated supervised ML models on structured business datasets and integrated their outputs into team reporting workflows, improving consistency and reproducibility.

Data Science Intern | DevelopersHub Corporation Jun 2025 – Jul 2025 | Remote

- Built Power BI dashboards tracking 8+ business KPIs, developed ML classification models, and automated data preprocessing with Python, reducing manual effort by about 15%.

PROJECTS

Nexora Knowledge Assistant | Production RAG System on AWS | GitHub Sep 2026 – Oct 2026
Python, FastAPI, LangGraph, Qdrant, BM25, Groq, PostgreSQL, Redis, Docker, GitHub Actions, AWS, DeepEval, LangSmith

- Built a RAG assistant over 10 company PDFs with hybrid retrieval (dense + BM25 + Reciprocal Rank Fusion) and cross-encoder reranking; raised hit@3 from 0.90 to 1.00 and MRR from 0.794 to 0.889 on a 30-question golden set.
- Profiled the pipeline stage by stage, found reranking took 78% of latency, and cut it from 7.4 s to 2.8 s with no loss in retrieval quality; added a role-aware Redis answer cache.
- Evaluated answers with DeepEval and an LLM judge (faithfulness 0.989, answer relevancy 0.933) and added a regression check that fails the build when a metric drops.
- Enforced role-based access control inside vector search (JWT, 5 roles), plus PII masking, prompt-injection and output guardrails; verified with 28 integration tests and a 29-case red-team suite.
- Shipped CI/CD: GitHub Actions (lint, type checks, 230 tests, 84% coverage) builds Docker images to Amazon ECR and deploys to EC2 through SSM with automatic health-check rollback; monitoring with Prometheus, Grafana and LangSmith.

AI Research Agent | Multi-Agent System | GitHub Jun 2026
Python, CrewAI, Groq, Tavily Search API, Pydantic, Gradio

- Built a 3-agent CrewAI pipeline (Researcher, Analyst, Writer) that searches the web with Tavily and writes a structured markdown report on any topic, with Pydantic validation of inputs and outputs and a Gradio web UI.

YouTube Comment Sentiment Analyzer | NLP + MLOps | GitHub Sep 2025 – Oct 2025
Python, NLP, Scikit-learn, FastAPI, MLflow, DVC, Docker, AWS EC2, JavaScript

- Trained a TF-IDF + Logistic Regression sentiment model (88% test accuracy on 3,700+ comments), tracked with MLflow and DVC, served as a Dockerised FastAPI on AWS EC2 and shown live in YouTube through a Chrome extension.

EDUCATION & CERTIFICATIONS

Bachelor of Science in Computer Science | University of South Asia 2021 – 2025 | Lahore
CampusX: Data Science Mentorship Program (6 months), AI Agents with CrewAI, AI Agents using Agno, GenAI using Ollama